

Mateo Alado

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EDUCATION

California State University, East Bay

August 2024 – December 2026

Bachelor of Science in Computer Science

Hayward, CA

- Distinctions: 2x AI Research Publications, HackHayward 2026 Winner, 3x Dean's List
- Coursework: Data Structures and Algorithms, Software Engineering, Operating Systems, AI, NLP

PROFESSIONAL EXPERIENCE

Software Engineer Intern (Agents)

June 2026 – August 2026

Lawrence Berkeley National Laboratory

Berkeley, CA

- Engineered a pipeline of 4 LangGraph agents for scientific literature ingestion and retrieval of 16,000+ publications
- Optimized retrieval time from 3+ minutes to under 5 seconds by building a PostgreSQL-backed GraphQL API
- Deployed agents on an HPC cluster using Slurm, scaling literature extraction and reducing processing time by 70%
- Redesigned UI by incorporating knowledge graph rendering with React, deploying to 450+ ALS beamline scientists
- Resolved a bottleneck in agent architecture by implementing a semaphore, enabling 25% more concurrent API calls
- Created 100+ end-to-end backend/agent tests with pytest, resulting in 95%+ total codebase test coverage

Software Research Engineer Intern

January 2026 – May 2026

California State University, East Bay

Hayward, CA

- Developed a FastAPI backend with context retrieval for data management, cutting request context size by 79.7%
- Built a script for automating data migration to a NoSQL database via cronjob, improving processing time by 60%
- Optimized a BFS adaptive graph traversal algorithm, achieving an 89% hit rate for LLM response accuracy
- First-authored an IEEE paper on GenAI applications in academic research for the GAISS 2026 conference

PROJECTS

Perplx - HackHayward 2026 Winner | TypeScript, Bun, Cloudflare Workers, Cloudflare D1, MCP

- Developed a terminal-native AI coding agent with MCP tooling using the Perplexity API for context-aware development and debugging
- Built a session-sharing infrastructure using Cloudflare Workers and D1 enabling real-time collaborative sessions via shared URLs
- Reduced cost per completed task to approximately \$0.10 through auto compaction and prompt caching

Publication Management Platform | Python, TypeScript, React, FastAPI, Neo4j, Supabase, Docker

- Built and deployed a full-stack RAG research platform using React and FastAPI for 20+ users to query and manage 100+ research publications and datasets
- Implemented persistent session context retrieval and user data storage with Supabase and Neo4j, cutting end-to-end latency to approximately 2 seconds
- Integrated vLLM and LiteLLM to support high-throughput local inference across 7 models from 3 LLM providers

Visual Speech Classifier | Python, PyTorch, NumPy, CNN, MediaPipe, OpenCV

- Created a real-time lip-reading classifier and pipeline using MediaPipe Face Mesh and OpenCV webcam capture to classify speech patterns for hearing-impaired individuals
- Achieved a 94% validation accuracy across 24 speech classes through iterative CNN optimization and refinement

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, Go

AI/ML: LLMs, AI Agents, LangGraph, LangChain, RAG, PyTorch

Backend: FastAPI, REST APIs, GraphQL, Flask, Node.js, Express

Database: PostgreSQL, MongoDB, NoSQL, Supabase, Neo4j

Frontend: React, React Native, Next.js, Tailwind, Vite

Developer Tools: Docker, GCP, AWS, Git, Linux, Slurm, Cursor, Claude Code